# Bhavesh Shrimali


**Contact Information**

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**Research Interests**

Computational and Applied Math; Numerical Analysis; High Performance Computing; Nonlinear Variational Problems; Scientific Machine Learning; Finite Elements

**Education**

**University of Illinois Urbana-Champaign** **4.0/4.0**

- MS and Ph.D., Computational Mechanics, 2015-present
  Advisor: Prof. Oscar Lopez-Pamies
- Minor: Computational Science and Engineering, 2016-present

**Indian Institute of Technology (IIT) Guwahati** **9.22/10.00**

- B.Tech, Civil Engineering, 2011-15
- *Ranked* 1st *in the department*

**Publications**

[2] **Bhavesh Shrimali** and William J. Parnell and Oscar Lopez-Pamies. A simple explicit model constructed from a homogenization solution for the large-strain mechanical response of elastomeric syntactic foams. *International Journal of Nonlinear Mechanics*, 2020 [**pdf**]

[1] **Bhavesh Shrimali** and Victor Lefèvre and Oscar Lopez-Pamies. A simple explicit homogenization solution for the macroscopic elastic response of isotropic porous elastomers. *Journal of the Mechanics and Physics of Solids*, 122:364 – 380, 2019 [**pdf**]

**Awards & Achievements**

- **Travel Fellowship**: Awarded to present my work at **WCCM-ECCOMAS, 2020** [*'20*]
- **Outstanding TA**: Ranked as an excellent TA for Structural Mechanics [*'17, '19*]
- **Institute silver medal**, IIT Guwahati: Ranked first in the graduating class [*'15*]
- **DAAD-WISE fellowship**: Awarded to approximately 150 Indian undergraduates to pursue a summer research internship in Germany [*'14*]
- **IET India Scholarship**: Awarded to about 170 students from select universities in India based on a nationwide scholastic exam [*'14*]
- **HKUST IIP**: Selected for the International Internship Program (IIP) to pursue a summer research internship at HKUST, Hong Kong [*'13*]
- **OPJEMS**: Among the top 50 students selected from all over India for the scholarship [*'13*]
- **Dhrishti 2012**: Among the *top 5* teams who proposed novel unconventional energy solutions for meeting India's future energy needs [*'12*]
- **Institute Merit Scholarship**, IIT Guwahati: Awarded annually to the top ranked student in each department based on the year round academic performance [*'12-'15*]
- **Indian National Olympiads**: Selected for the national level rounds in Mathematics, Physics and Astronomy [*'11*]
- **KVPY Fellow**: Ranked among 209 students from all over the country [*'10*]
- **NTSE Scholar**: Secured an All India Rank (AIR) of 324/1000 [*'07*]

**Technical Skills**

**Languages:** Python, C++, Julia, Fortran

**Finite Element Libraries:** FEniCS, Firedrake, scikit-FEM, GridAP.jl

**Distributed Computing:** MPI, OpenMP, CUDA, OpenCL

**Miscellaneous:** Bash, Git, PyCUDA, pybind11, **Legate-NumPy**

## Computational Projects

**Solving Nonlinear PDEs using Deep Learning** *2020-*

Implementing a coupled framework with finite elements (in space) and deep neural networks (in time) to solve nonlinear PDEs arising in viscoelasticity and fracture. The overarching goal is to develop an incremental variational framework that can describe and predict the nucleation and propagation of fracture in heterogeneous (and dissipative) material systems when subject to arbitrary loading in space and time.

**Numerical Homogenization of Multiscale PDEs** *2016-2019*

*Guide: Oscar Lopez-Pamies, (CEE) UIUC* [**code**]

Implemented parallel, stable and robust finite element (FE) discretizations to solve second order nonlinear variational problems on periodic manifolds arising in mechanics. Recent work involved coupling these with fifth order *time*-explicit-integration algorithms to solve PDEs in nonlinear viscoelasticity. All the algorithms were implemented in the open source FE framework **FEniCS** and the resulting system of nonlinear algebraic equations were solved via canned solvers in **PETSc**. The numerical schemes were validated against a proposed analytical solution and were demonstrated to be optimally convergent (in time and space).

**Non-Conforming Finite Elements for Biharmonic Problems** *2018-*

*Guide: Oscar Lopez-Pamies, (CEE) UIUC* [**ongoing**]

Implemented and solved the biharmonic plate-equation describing the small deformation response of perforated Kirchhoff plates. A non-conforming FE discretization using the optimally convergent Morley-finite element basis coupled with Nitsche's method was used to impose free boundary conditions. The FE assembly was performed using a vectorized FE code written in **scikit-FEM** which builds on top of NumPy. The future goal is to accelerate the FE code on GPUs via the distributed computing features offered by NVIDIA's **Legate-NumPy**

**Multigrid Finite Elements in Hyperelasticity** *2018*

*Guide: Luke Olson, (CS) UIUC* [**report**]

Implemented a vectorized FE code (in pure NumPy) with multigrid preconditioners/multilevel solvers from **PyAMG** to simulate the response of hyperelastic materials over a large range of physical properties. This work was done in collaboration with Tejaswin Parthasarathy. We demonstrated that the multilevel solvers in PyAMG when tuned appropriately, can compete with their direct counterparts, e.g. SUPERLU in `scipy.sparse.linalg.spsolve`, in both speed and accuracy, even for small problems ($N \sim$10k dofs)

## Teaching

- CEE 598: Constitutive Modeling
- CEE 471: Structural Mechanics [Excellent TA.]
- CS 125: Introduction to Computer Science
- CSE 551: Finite Element Methods
- CS 357: Numerical Methods

## Research Internships

**Technische Universität Braunschweig** [*Summer '14*]

*Advisor: Prof. Dr. Klaus Thiele*

I worked on linear perturbation analysis of a cable stayed bridge to identify the dominant eigen-modes. Subsequently, based on a FE analysis of the galloping cables, I proposed an optimum location of an eddy current damper that resulted in maximum reduction of the dominant eigen value.

**Hong Kong University of Science and Technology** [*Summer '13*]

*Advisor: Prof. Christopher K Y Leung* I worked on tuning the proportion of elastomeric and steel fibres in high performance concrete to enhance the post peak softening response. I also worked on validating a FE analysis of a notched beam subject to 3-point bending with the in situ experiments.

## Courses

Fast Algorithms and Integral Equations, Multigrid Methods, Nonlinear Finite Elements, Generalized Finite Element Method, Numerical methods for PDEs, Computational Inelasticity